Math 150A Section 8 and 16
Spring 2025
Practice Exam II
March 06, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): ____________________

Student ID: ____________________

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	6	
4	10	
5	10	
6	10	
7	7	
Total:	63	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write

   (a) if $f''(3) = 0$ then $f(x)$ has an inflection point at 3

   (b) the function $f(x) = |x|$ has no critical points

   (c) if $f(x)$ is continuous on the interval $[0, 1]$, then it has a has both absolute maximum and absolute minimum values on $[0, 1]$

   (d) if $f'(x) < 0$ for all x, then $f(x)$ can have at most one root

   (e) if $f(0) = 0$ and $f(1) = 60$ then $f'(x) = 60$ for some value of x in $(0, 1)$

2. (10 points) Water is leaking out of an inverted conical tank at a rate of $10,000\ \text{cm}^3/\text{min}$ at the same time water is being pumped into the tank at a constant rate. The tank has a height of 6 m and the diameter at the top is 4 m. If the water level is rising at a rate of 20 cm/min when the height of the water is 2 m, what is the rate at which water is being pumped into the tank?

3. (6 points) In class, we learned a list of critical properties to consider when sketching a function. Suppose that $f(x)$ is a function which simultaneously satisfies all of the following:

(A) the domain is $\mathbb{R}$

(B) there is a y-intercept at 0; there are x-intercepts at 0 and $\pm 3\sqrt{3}$

(C) the function is odd

(D) there are no horizontal or vertical asymptotes

(E) the function is increasing on $(-\infty, -1)$ and $(1, \infty)$, and is decreasing on $(-1, 1)$

(F) it has a local maximum $f(-1) = 2$ and local minimum $f(1) = -2$

(G) the function is concave upward on $(0, \infty)$, concave downward on $(-\infty, 0)$ with an inflection point at $x = 0$

Using these properties, we should be able to produce a remarkably good sketch of $f(x)$, even though we don't even know what $f(x)$ is! Create a sketch of $f(x)$ which simultaneously satisfies all the above properties.

4. (10 points)

   (a) State the Mean Value Theorem

   (b) State the Extreme Value Theorem

5. (10 points) For each of the following, provide an example or explain why it does not exist.

   (a) a function $f(x)$ with a critical point at 0 but neither a local max nor local min at $x = 0$

   (b) a function $f(x)$ which satisfies $f''(1) = 0$ but which does not have an inflection point at $x = 1$

   (c) a nonconstant function $f(x)$ whose derivative is always zero

   (d) a function with an inflection point at $x = 0$

   (e) a differentiable function with three x-intercepts, whose derivative only has one x-intercept

6. (10 points) If $1,200$ cm^2 material is available to make a box with a square base and an open top, find the largest possible volume of the box.

7. (7 points) Consider the function

$$f(x) = x^4 + x^2 - 1$$

(a) show that $f(x)$ has only one critical point

(b) explain why the Mean Value Theorem says $f(x)$ cannot have more than two roots

(c) calculate the value of $f(-1)$, $f(0)$ and $f(1)$

(d) explain why the Intermediate Value Theorem says $f(x)$ has at least two real roots (and therefore precisely two roots)